PaperCast: An Interactive, Evidence-Grounded Audio Learning Assistant

Project Abstract

by

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ABSTRACT

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Academic papers and technical documents are a core part of higher education, research, and professional learning, yet their length, complexity, and visually dense content can make them difficult to consume efficiently. Audio-based learning offers a hands-free, accessible alternative and advances in generative AI now make it possible to convert static documents into fluent, conversational audio, leading to growing adoption of document-to-audio tools.

Existing document-to-audio tools largely emphasize content generation rather than the complete learning experience. Listeners cannot interrupt playback to ask questions, request a different explanation, or navigate directly to a topic. Studies on similar AI tools report hallucination rates of 13-40%, so listeners may not know when a claim is inaccurate. Visual content is also hard to access through audio, and explanations don't adapt to what the listener understands.

In this project, we propose PaperCast, an evidence-grounded agentic audio learning assistant for academic documents. A Plan, Execute, Evaluate, and Retry orchestrator coordinates document extraction, retrieval, audio generation, and verification. Users can listen to conversational audio, ask questions, request simpler or deeper explanations, navigate by topic, and trace claims to source content. Scientific integrity checks verify generated information against the source, while adaptive learning identifies knowledge gaps, guides questioning, and addresses misconceptions. PaperCast also makes figures, tables, and equations accessible through audio. The expected outcome is a reliable, adaptive, accessible and interactive audio learning experience for understanding complex academic content.